Turkish-CBE: Evaluating Logical Reasoning Accuracy and Morphosyntactic Robustness in Large Language Models

A Controlled Turkish–English Benchmark with Correctness-Aware Robustness Evaluation

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Abstract

Accuracy on a single language and a single formulation is an incomplete account of logical reasoning reliability: a model may answer correctly only under one grammatical realization of a problem, or may appear stable merely by repeating a biased answer. We present Turkish-CBE, a deterministic, reproducible benchmark that renders the same latent ordering problem as matched Turkish and English prompts and as answer-preserving Turkish presentation transformations (constituent order, converse constructions, case/possessive alternatives), with exact symbolic answers, controlled entity counts and query distances, and split-level separation of latent orders. A correctness-aware evaluation decomposes paired outcomes into robust correctness, shared error, and presentation dependence on declared denominators, distinguishing correct consistency from constant-response invariance. We report an exploratory evaluation of GPT-4o and the Qwen2.5 instruct family (0.5B–14B; 16,618 matched prompts per model in a supplemental quantized sweep). Sub-7B models are near-constant responders whose perfect invariance carries no competence; discrimination emerges in English before Turkish; and a truth-preserving Turkish constituent-order alternative (P3) costs the 14B model 38.1 points of query-matched paired accuracy (cluster-bootstrap 95% CI [34.4, 41.8]), echoing GPT-4o's pilot collapse to 36.9%. A completed two-reviewer linguistic review of 360 sampled stimuli finds unanimous semantic equivalence, excludes one construction on grammaticality—after which the P3 effect persists at 40.8 points—and records contested naturalness. Reasoning reliability should therefore be evaluated jointly over accuracy and meaning-preserving presentation robustness; confirmatory frozen-test evaluation remains future work.

Keywords: large language models; logical reasoning evaluation; Turkish; morphosyntax; cross-lingual evaluation; robustness; paired evaluation design; reproducibility; benchmark construction; chance-corrected agreement

1 Introduction

A logical conclusion can remain unchanged when its premises are reordered, a relation is expressed through its converse, or a language uses a different grammatical realization. A model that changes its answer under such transformations presents a different evaluation problem from a model that consistently reaches the wrong conclusion, and both differ from a model that appears stable only because it repeats a biased answer. Aggregate accuracy on a single linguistic realization obscures all three distinctions: equal scores can conceal failures on different instances, and high agreement can arise without any competence at all. Behavioral tests and consistency evaluations therefore complement conventional task-level scores (Ribeiro et al., 2020; Elazar et al., 2021).

Turkish offers an unusually well-controlled setting for making these distinctions measurable. It expresses grammatical relations through productive suffixation and morphophonological alternations, and permits constituent-order choices whose discourse effects a literal translation does not capture; computational descriptions of Turkish morphology formalize the interaction between morphotactics and phonological realization (Oflazer, 1994). These properties supply a family of *answer-preserving* transformations (premise reordering, converse constructions, constituent-order alternatives, and case or possessive variants) that can be applied to the same underlying problem. We treat Turkish as a source of controlled variation, not as a language presumed intrinsically harder for logical reasoning.

Existing resources do not answer how equivalent Turkish formulations affect transitive ordering inference. General logical reasoning benchmarks (Yu et al., 2020; Liu et al., 2020; Han et al., 2024) vary surface form and content together; multilingual reasoning evaluations (Shi et al., 2023) compare independently sampled items across languages; Turkish morphological-competence and minimal-pair benchmarks (Ismayilzada et al., 2025; Başar et al., 2025; Başar and Bisazza, 2026) intentionally manipulate acceptability rather than holding the answer fixed; and paraphrase-consistency and metamorphic evaluations (Elazar et al., 2021; Segura et al., 2016) motivate, but do not provide, a matched bilingual ordering design with exact answers. Competence-Based Evaluation (CBE) treats correctness across presentations of a parameterized problem family as a behavioral requirement (Jiang and Zhai, 2026); we adopt this framing and extend it to Turkish–English matching, morphosyntactic interventions, lexical measurements, and query-distance controls.

Turkish-CBE renders each latent strict total order as matched Turkish and English prompts and as controlled Turkish presentation transformations, with exact symbolic answers, explicit entity-count and query-distance factors, and split-level separation of latent orders. On top of this design, a correctness-aware analysis decomposes paired outcomes on jointly valid support into robust correctness (R), shared-wrong agreement (D), and presentation dependence (U), reported together with raw invariance, a chance-corrected agreement coefficient, valid-pair coverage, and strict all-planned robustness (Section 4). These quantities are complementary: invariance alone cannot separate correct consistency from stable error, and accuracy alone cannot reveal presentation dependence.

The framework is exercised by completed exploratory experiments. A registered pilot evaluates Qwen2.5-0.5B-Instruct and `gpt-4o-2024-08-06` on matched prompts (Section 7); a supplemental quantized local sweep evaluates four Qwen2.5 scales (0.5B–14B) on 16,618 matched prompts each across all nine relation families (Section 7.7); and a two-reviewer Turkish linguistic review of 360 sampled stimuli was completed after these runs (Section 7.8), certifying semantic equivalence unanimously, excluding one construction on grammaticality, and documenting contested naturalness.

This article makes three principal contributions:

1. **A controlled bilingual reasoning benchmark.** A deterministic, reproducible framework that holds the latent problem, gold answer, names, and query geometry fixed while varying language and Turkish presentation, supporting more controlled comparisons than independently sampled multilingual test sets. The design does not claim to eliminate every linguistic or tokenization confounder; it makes the remaining ones explicit and measurable.

2. **Correctness-aware presentation robustness.** A denominator-aware paired decomposition that separates correctness from response stability. Its components are related to existing agreement and paired-outcome methods rather than constituting a new statistical theory; their joint presentation is what carries the evaluation content. The constant-response behavior of the smaller evaluated models provides a concrete empirical demonstration that high response invariance does not indicate reasoning competence.

3. **An empirical characterization of linguistic presentation sensitivity.** Within the evaluated models, datasets, and settings: model-scale differences in discriminative accuracy, English–Turkish performance differences, strong sensitivity to the P3 constituent-order presentation, complexity-dependent degradation, naming-condition effects, and the distinction between correct consistency and constant-

response invariance. These are findings about the evaluated systems, not generalizations to all large language models.

Turkish-CBE addresses five research questions. RQ1 concerns the paired Turkish–English accuracy difference; RQ2 sensitivity to answer-preserving Turkish presentations; RQ3 the separation of entity count from query distance and query type; RQ4 associations among relation vocabulary, corpus frequency, tokenization, and predicted-answer bias; RQ5 whether reasoning prompting or optional adaptation improves correct consistency as well as accuracy. The completed exploratory evidence bears on RQ1–RQ3 (and supplies RQ4 diagnostics), while confirmatory answers to all five—and any RQ5 adaptation effect—are reserved for the preregistered frozen-test design (Section 9.2).

2 Related Work

2.1 Logical reasoning benchmarks and controlled generation

Natural-language reasoning benchmarks differ in their source material and the inferential structures they expose. ReClor and LogiQA evaluate reasoning-oriented reading comprehension (Yu et al., 2020; Liu et al., 2020). FOLIO connects natural-language arguments to first-order logical representations (Han et al., 2024). Their task diversity is valuable, but a change in the language or surface form of an item need not preserve all other evaluation conditions. Turkish-CBE instead fixes a narrow ordering family so that the latent question and its answer can be held constant across presentations.

Generated benchmarks provide complementary structural control. CLUTRR tests relational reasoning from text (Sinha et al., 2019); RuleTaker studies reasoning over natural-language rule bases (Clark et al., 2020); and ProofWriter adds generated implications, proofs, and abductive statements (Tafjord et al., 2021). PrOntoQA supports formal analysis of reasoning chains (Saparov and He, 2023). These precedents show why explicit structure and an oracle are useful. Our contribution is not the invention of generated reasoning data; it is their use within a matched Turkish presentation design, with input size and latent proof distance separately recorded.

2.2 Robustness, metamorphic relations, and consistency

Metamorphic testing specifies relations between outputs under controlled input transformations (Segura et al., 2016). CheckList applies behavioral test design to language technologies (Ribeiro et al., 2020). ParaRel studies consistency across paraphrased factual queries (Elazar et al., 2021); factual cloze consistency is distinct from deductive ordering, but it motivates separating correctness from stability. CBE combines a scalable latent family with an explicit answer-preservation requirement (Jiang and Zhai, 2026). We retain that requirement and add Turkish-specific interventions and missing-response accounting.

Controlled perturbations in arithmetic also motivate careful interpretation. GSM-Symbolic evaluates variation in mathematically structured problems (Mirzadeh et al., 2025). Its task differs from ordering, so neither its numerical findings nor a particular mechanistic explanation is imported here. A presentation effect can reflect parsing, retrieval, lexical priors, or inference; behavioral inconsistency alone does not distinguish those causes.

2.3 Multilingual reasoning and Turkish linguistic evaluation

MGSM investigates multilingual chain-of-thought arithmetic reasoning (Shi et al., 2023). Turkish is not one of its evaluated languages, and arithmetic success cannot validate Turkish ordering templates. A multilingual ordering benchmark must therefore establish its own linguistic equivalence and matched sampling.

Turkish morphology has long motivated explicit computational modeling (Oflazer, 1994). Morphological compositionality experiments expose limitations in combining morphemes productively (Ismayilzada et al., 2025). TurBLiMP evaluates Turkish minimal-pair acceptability (Başar et al., 2025), whereas morphology-aware syntax evaluation examines how surface factors can bias judgments (Başar and Bisazza, 2026). Those tasks are complementary to ours: minimal-pair grammaticality contrasts often intentionally change acceptability, while Turkish-CBE requires grammatical alternatives that preserve truth and the gold answer. Native-speaker adjudication remains necessary despite deterministic rendering.

2.4 Tokenization, frequency proxies, and reasoning interventions

Tokenizer quality and language coverage can affect multilingual model behavior (Rust et al., 2021). Token-length inequalities also matter for access costs and processing budgets (Ahia et al., 2023; Petrov et al., 2023). These observations justify measuring tokenization rather than assuming equal cost for matched text. They do not establish a causal tokenization explanation for a logical error. Corpus frequency is likewise a proxy with explicit genre, annotation, and segmentation limits; our independent resource is the locked BOUN treebank release (Türk et al., 2022).

Chain-of-thought prompting and self-consistency are distinct interventions (Wei et al., 2022; Wang et al., 2023). Self-consistency aggregates sampled responses to one input; presentation invariance compares different inputs that preserve the same answer. Moreover, generated explanations need not faithfully report the causes of an answer (Turpin et al., 2023). Parameter-efficient adaptation with LoRA (Hu et al., 2021) and group-relative policy optimization (Shao et al., 2024) motivate optional training arms, but their success elsewhere provides no evidence that our unrun adapters improve Turkish reasoning.

3 Methodology

3.1 Latent structure, oracle, and query geometry

Let a latent instance be a strict total order

$$z = (e_{\sigma(1)} \prec e_{\sigma(2)} \prec \cdots \prec e_{\sigma(n)}), \qquad |E| = n. \tag{1}$$

The frozen size grid is $n \in \{4, 8, 16, 32, 64, 128, 256, 512, 1024\}$. The generator draws a permutation of unique entities and emits the $n - 1$ adjacent comparisons. A language-independent rank map $r_z : E \to \{1, \ldots, n\}$ defines the exact answer to a forward query:

$$y^*(z, a, b, \prec) = \mathbf{1}\{r_z(a) < r_z(b)\}. \tag{2}$$

The inverse relation reverses the inequality. A separate graph check validates the chain and its entailed comparisons. Stable, versioned seed derivation separates naming, permutation, premise order, and query selection, so introducing a new presentation condition does not alter previously defined latent instances.

Endpoint queries compare the first and last positions. Internal queries sample two positions strictly inside the chain. Distance-controlled queries fix

$$d = |r_z(a) - r_z(b)|, \tag{3}$$

with both positions internal. Thus their feasible distances satisfy $1 \le d \le n - 3$; the retained power-of-two grid is intersected with that range. This is narrower than merely requiring $d < n$. At $n = 4$, an internal pair necessarily has $d = 1$, so internal and distance-controlled conditions can share geometry. Such repetitions are not independent evidence.

A longer endpoint chain does not prove deeper inference. The source and sink can be identified by their incident-edge signatures under the chain assumptions, avoiding a full transitive derivation. Entity count measures input and presentation load; d measures latent graph distance, not observed model computation. We therefore retain endpoint and internal/distance strata and do not treat their pooled score as an inference-depth curve.

Table 1: Turkish presentation factors. P0 denotes the canonical form. The general matched equivalent condition is retained separately from isolated ablations.

Factor	Operation	Interpretation control
P0	Canonical grammatical realization	Baseline; identical latent query.
P1	Reorder premises and names list	Preserve words and predicates.
P2	Swap arguments and use inverse relation	Preserve the asserted comparison.
P3	Alternative constituent construction	Truth preservation does not imply identical discourse preference.
P4	Valid case/possessive morphology alternatives	Label additional syntax/lexical changes explicitly.
P5	Combine several operations	Compound perturbation; no pure morphology attribution.
P6	Optional controlled negation	Excluded from the frozen core; requires strict-order proof.

3.2 Language realization and equivalent presentations

A renderer $\pi_{\ell,p,v,c}$ maps a latent query to language ℓ, presentation p, relation vocabulary v, and name control c. Its formal contract is

$$\mathcal{N}(\pi_{\ell,p,v,c}(z)) = \mathcal{N}(z), \qquad y^*(\pi_{\ell,p,v,c}(z)) = y^*(z), \tag{4}$$

where $\mathcal{N}$ is the closed-grammar logical normal form. Converse realization swaps arguments and uses the inverse predicate; for example, $a \prec b$ is realized as either a forward statement about a or an inverse statement about b. Premise order and names-list order are independent surface factors.

The inventory contains nine relations from spatial, temporal, comparative, and ranking families. Spatial relations have explicitly linear positional interpretations. Turkish realization selects genitive/possessive or ablative constructions as appropriate, handles proper-name apostrophes, and applies vowel and consonant harmony. The reference to Turkish morphological theory (Oflazer, 1994) motivates these constraints; it does not certify our implementation. Table 1 separates operational transformations from linguistic claims.

3.3 Semantic validation and human adjudication

Automatic validation checks distinct entities, chain consistency, premise truth, query distance, normal-form equivalence, and oracle agreement. Rendering and parsing are partly based on shared linguistic code, creating a common-mode failure risk. Passing a round trip establishes a contract within the implemented grammar; it is not an independent proof of naturalness or unrestricted Turkish equivalence.

Two independent Turkish-proficient reviewers must judge grammaticality, naturalness, semantic equivalence, and morphology, with disagreements resolved before an approved review record is issued. The operational large-run gate requires at least 100 distinct reviewed Turkish prompts, at least 20 in each selected relation–presentation–name template cell, and coverage of every requested size and query type. This is sampled review, not exhaustive review of every cross-cell. Existing automatic checks and an unfilled review sheet do not constitute human approval. A review of this form was executed for the evaluated prompt population; its instrument, predeclared inclusion policy, and outcomes are reported in Section 7.8. Work on Turkish minimal pairs reinforces the need to separate grammatical evaluation from logical labels (Başar et al., 2025; Başar and Bisazza, 2026).

3.4 Bilingual matching, names, and split integrity

Turkish and English records share the latent-order identifier, query target, query direction, gold label, and presentation plan. The `native_shared` control uses the same native Unicode spelling in both

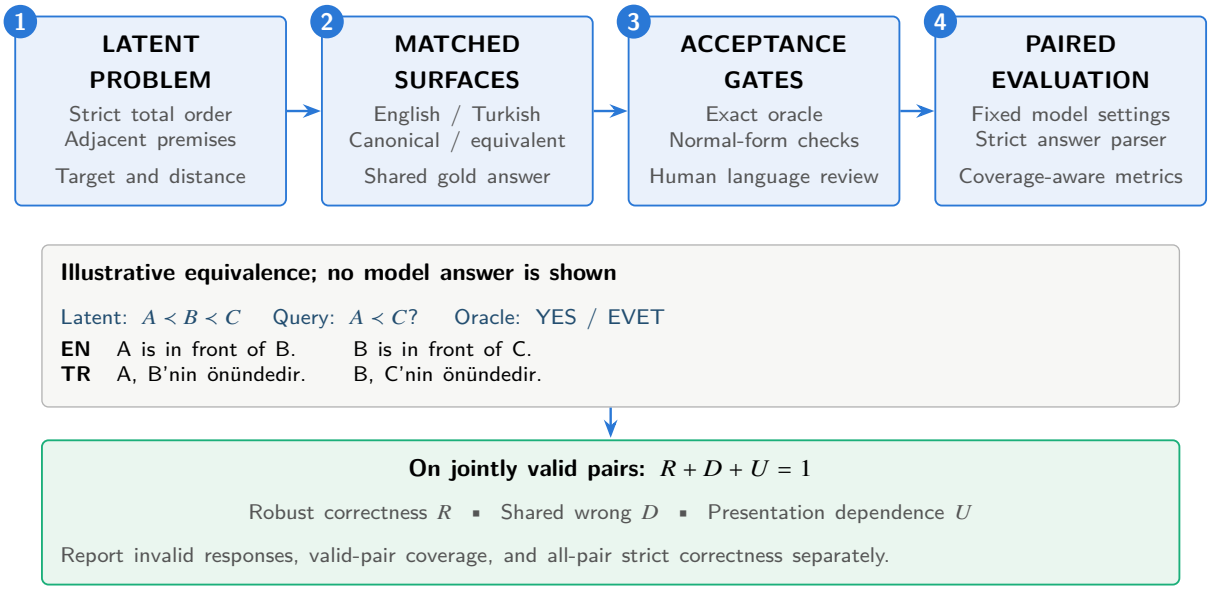


Figure 1: Turkish-CBE architecture and an illustrative bilingual realization. The diagram describes the evaluation design, not observed model accuracy.

languages. `en_transliterated` changes only the English labels, retaining native Turkish spelling and morphology. `symbolic_shared` assigns identical symbolic labels in both languages. Consequently, the transliterated arm intentionally changes naming together with language; it is a naming-sensitivity control, not a pure language contrast. The shared-name arms support the cleaner matched comparison. They do not remove all differences in tokenizer vocabulary, instructions, or response labels. All derivatives of a latent order are assigned to one split before rendering; changing the vocabulary cannot move the same instance from training into test.

The upstream generator supports several entity inventories, but the frozen paired bank analyzed here uses the Turkish-name source inventory before alias expansion; a fully crossed synthetic-name experiment is not established by these counts. The unit of independence is the source order, not a prompt record or comparison cell. The total-order family is structurally isomorphic under renaming: exact train/test instance separation is not separation of abstract reasoning algorithms or a guarantee against external pre-training contamination. The current frozen grid crosses three name controls, nine relations, and nine language/presentation conditions: two English conditions and seven Turkish conditions (canonical, general equivalent, and P1–P5). This produces $3 \times 9 \times 9 = 243$ prompt identities for each source query. Optional negation is not silently included in that expansion.

4 Correctness, Invariance, and Statistical Inference

4.1 Denominators and the paired outcome decomposition

For N planned matched presentation pairs, let $a_{ik} \in \{0, 1, \perp\}$ be the strictly parsed answer to presentation $k \in \{0, 1\}$, where $\perp$ denotes invalid or missing output. Marginal correctness is $C_{ik} = \mathbf{1}\{a_{ik} = y_i^*\}$, with invalid answers wrong. Define $V_i = \mathbf{1}\{a_{i0} \neq \perp,\ a_{i1} \neq \perp\}$ and $M = \sum_i V_i$. The marginal accuracies and valid-pair coverage are

$$\widehat{A}_k = \frac{1}{N} \sum_{i=1}^{N} C_{ik}, \qquad \hat{v} = \frac{M}{N}. \tag{5}$$

For $M > 0$, the paired CBE quantities use exactly the same jointly valid support:

$$\widehat{R} = \frac{1}{M} \sum_i V_i \, C_{i0} C_{i1}, \tag{6}$$

$$\widehat{D} = \frac{1}{M} \sum_i V_i \, (1 - C_{i0})(1 - C_{i1}), \tag{7}$$

$$\widehat{U} = \frac{1}{M} \sum_i V_i \, \mathbf{1}\{C_{i0} \neq C_{i1}\}. \tag{8}$$

Consequently,

$$\widehat{R} + \widehat{D} + \widehat{U} = 1, \qquad \widehat{A}_{\text{valid}} = \widehat{R} + \widehat{U}/2, \qquad \hat{I} = \widehat{R} + \widehat{D}. \tag{9}$$

Here $\widehat{A}_{\text{valid}}$ is the mean of the two accuracies *on jointly valid pairs*. It is not automatically $(\widehat{A}_0 + \widehat{A}_1)/2$ on the full planned sample. Binary gold labels shared within each pair imply that D is shared-wrong agreement, whereas U is presentation dependence.

For $N > 0$ and $M > 0$, strict robust correctness retains all planned pairs:

$$\widehat{R}_{\text{strict}} = \frac{\sum_i V_i \, C_{i0} C_{i1}}{N} = \hat{\nu} \, \widehat{R}. \tag{10}$$

The equality exposes selective coverage directly. It does not assert that marginal accuracy equals coverage times valid-pair accuracy: pairs with one valid, correct answer can still contribute to a marginal. Both response-level invalid rates and pair coverage must accompany every conditional profile. If $M = 0$ and $N > 0$, strict robustness is zero while conditional R is undefined; the product identity is used only when R is defined. Empty conditional denominators are undefined, not zero.

4.2 Chance correction and its interpretation

At a fixed complexity and declared model/condition stratum, define q as the pooled YES/EVET rate over the $2M$ jointly valid answers. The retained CBE reference is

$$I_e = q^2 + (1 - q)^2, \qquad \kappa_{\text{CBE}} = \frac{\hat{I} - I_e}{1 - I_e}. \tag{11}$$

Constant outputs make $I_e = 1$, so the coefficient is undefined. High raw agreement from a constant answer is therefore not evidence of competence. The chance correction is related to the general idea of agreement beyond chance (Cohen, 1960), but its pooled reference is not identical to the separate-marginal reference in conventional Cohen kappa. Appendix A derives the difference. Neither reference identifies an internal reasoning mechanism or proves stochastic independence.

A valid-pair mean accuracy alone bounds but does not determine robustness. From the nonnegative outcome masses,

$$\max(0, \, 2A_{\text{valid}} - 1) \leq R \leq A_{\text{valid}}. \tag{12}$$

Likewise, invariance cannot identify the division between correct and shared-wrong agreement. These algebraic observations explain why accuracy, R, D, U, coverage, and chance correction should be presented jointly rather than ranked as interchangeable notions of reliability.

4.3 Paired inference, dependence, and multiplicity

The primary cross-language accuracy effect is

$$\widehat{\Delta}_{\text{TR}-\text{EN}} = N^{-1} \sum_i (C_{i,\text{TR}} - C_{i,\text{EN}}), \tag{13}$$

using identical planned coordinates and counting invalid answers wrong. Effects are evaluated within declared model, size, query-type/distance, relation, presentation, and name-control strata. Any pooled

summary must specify whether it weights derived queries, latent orders, or size levels; those are different estimands. Paired McNemar inference (McNemar, 1947) compares discordant correctness outcomes when matched units are independent. The exact conditional version uses a binomial reference on discordant pairs. Repeated queries or presentations from one latent order violate ordinary row-level independence; such rows require cluster-aware uncertainty rather than an unqualified McNemar p-value.

Confidence intervals use 2,000 matched latent-order bootstrap replicates, preserving every dependent query/presentation record and resampling within declared size strata. This extends the resampling principle (Efron, 1979) to the clustered design (Field and Welsh, 2007). Wilson intervals (Wilson, 1927) are useful binomial summaries for R only under an appropriate independent-Bernoulli interpretation; when rows share latent orders, dependence-aware bootstrap intervals carry the inferential interpretation. Intervals are pointwise, not simultaneous bands.

Holm correction (Holm, 1979) applies within predeclared model/design/metric families, with missing planned hypotheses retaining their correction slots. The implementation is not study-wide familywise control across every size, relation, distance, and intervention. Confirmatory families must be specified before outcomes are observed. For base–adapter comparisons, all-planned accuracy, strict R, and coverage are primary; common-four-valid differences in conditional R, invariance, and kappa are explicitly labeled sensitivity analyses and report both arms' valid support.

4.4 Lexical associations and identifiable effects

For each relation direction, the predicted YES/EVET rate is the number of parsed affirmative answers divided by the number of parsed responses in that direction. Its denominator, planned gold-label balance, and invalid rate must be shown together. This descriptive rate may use lone valid responses; it is distinct from the pooled q in Equation (11), which uses jointly valid pairs only. A direction-specific accuracy change and invariance change need not agree.

Frequency–accuracy and tokenization–invariance analyses were executed on the pilot outcomes (Section 7) but remain unidentified there, because a single relation family supplies at most two lexical units per cell. Spearman association (Spearman, 1904) can summarize monotone relations, but a handful of repeated lexical units does not create independent observations through prompt replication. A multivariable specification may include model, relation family, $\log_2 n$, query distance, and prompt length, with uncertainty clustered by latent order and appropriate attention to lexical reuse. Saturated model-by-vocabulary effects may absorb all between-term frequency variation; unsupported or collinear fits must remain unavailable, and the retained pilot outputs record them as such rather than as coefficients.

We distinguish a total paired language contrast from an estimate adjusted for token length. Length may lie on a pathway by which language changes the presented input, so adjustment changes the question rather than automatically correcting a nuisance. These are descriptive associational analyses, not identification of a causal frequency or morphology effect. No lexical-association coefficient is promoted to a finding in the current article; the executed pilot specifications are retained with their explicit unidentifiability statuses.

5 Experimental Design and Reproducibility

5.1 Research questions and registered contrasts

Table 2 connects each research question to a planned estimand. Four-cell balancing crosses query direction and gold answer. Frozen test strata contain 200 source orders per size and 50 examples per direction–gold cell; smaller pilot/development/training strata use approximately balanced 12/13 cells where 50 cannot be split equally. Derived presentations retain those source labels rather than creating additional independent cases.

The recorded split generation seeds are 401 (training), 402 (development), 403 (pilot), and 404 (test), with presentation seed 431. Seed derivation remains versioned and namespaced. The experiment

Table 2: Research questions and the measurements needed to answer them. Exploratory estimates of the RQ1–RQ3 readouts (and RQ4 diagnostics) exist from the completed pilot and supplemental evaluations (Sections 7–7.8); confirmatory frozen-test estimates and the RQ5 adapter contrast remain pending.

RQ	Matched contrast	Required readout
RQ1	English vs. Turkish at fixed latent query, names, and presentation	Paired accuracy difference, invalid rates, coverage.
RQ2	Turkish canonical vs. equivalent/P1–P5	R, D, U, I, κ, strict R, support.
RQ3	Size and distance within each query type	Separate n and d profiles; endpoint/internal contrast.
RQ4	Relation terms and directions on identical instances	Frequency/token features, predicted YES rate, accuracy and invariance jointly.
RQ5	Direct vs. reasoning; base vs. trained adapter	Matched effects, common support, usage and compute budgets.

matrix separates English canonical/equivalent, Turkish canonical/equivalent, cross-language canonical, cross-language equivalent, Turkish ablations, and endpoint-to-interior contrasts. The last changes the queried proposition and may change the gold answer; it is a query-complexity comparison, not an answer-preserving presentation pair. We therefore do not compute presentation invariance by blindly pairing endpoint and internal answers as if they answered the same question.

5.2 Prompts, parsers, and generation budgets

The direct Turkish instruction asks the model to use the supplied premises and return only `EVET` or `HAYIR`; the English control requests `YES` or `NO`. A strict language-specific parser accepts only the specified form, records wrong-language outputs separately, and rejects contradictory or extra answers. It does not search arbitrary response text for a convenient label. Reasoning conditions instead require an unambiguous final-answer marker, with correctness determined by that answer rather than by the explanation.

A run must record immutable model and tokenizer identities, provider/serving revision, sampling parameters, seed where supported, system/user prompts, context and output budgets, reasoning setting, token usage, truncation, and cost status. Model unavailability cannot silently select a replacement. Planned tokens are not billed usage, and isolated-string token counts are not chat-template counts. Context failures, missing completions, and invalid responses stay in planned-sample denominators.

Dry-runs estimate call counts before a small real pilot. Full evaluation requires an approved pilot; large evaluations, including pilots or training evaluation exceeding 2,000 planned generation calls, also require dataset- and scope-bound human linguistic review. Checkpointing and append-only raw records support resumption without overwriting model evidence. An identical-prompt repeat control should accompany stochastic or provider-nondeterministic settings, so cross-presentation disagreement is not automatically attributed to linguistic variation. This is a prospective control, not a reported experiment. A local journal establishes recorded lineage, not independent provider attestation. Two registered configurations (Qwen2.5-0.5B-Instruct and `gpt-4o-2024-08-06`) were enabled by explicit user selection and executed under this protocol (Section 7); the supplemental quantized sweep reuses the frozen prompts, message construction, and strict parser outside the registered provider set and is labeled accordingly (Section 7.7). The execution-gate receipts for runs above the declared call threshold have not been issued, so no run in this article carries confirmatory authorization.

5.3 Reasoning and optional training interventions

Direct answering, open stepwise prompting, structured normalization/direction/transitivity prompting, and provider reasoning-effort controls are distinct arms. Their latent queries, gold labels, and sampling conditions should be held constant where possible. Output budgets must be sufficient and recorded;

Table 3: Verified dataset inventory. Orders are source chains; queries and presentations are dependent derivatives. Presentation counts describe the frozen design, not completed model calls.

Split	Orders	Queries	Presentations	Pairing cells
Training	450	3 550	862 650	1 197 450
Development	450	3 550	862 650	1 197 450
Pilot	300	1 900	461 700	634 500
Frozen test	1 800	14 200	3 450 600	4 789 800
Total	3 000	23 200	5 637 600	7 819 200

changes in budget can otherwise confound interpretation. CoT and sampled self-consistency (Wei et al., 2022; Wang et al., 2023) motivate interventions but are not the same as invariance across equivalent presentations. Final-answer improvements do not establish explanation faithfulness (Turpin et al., 2023).

Optional training compares a base model, SFT-LoRA, paired-presentation SFT, SFT with a consistency term, correctness-only GRPO, and correctness-plus-consistency GRPO. LoRA supplies a parameter-efficient adaptation mechanism (Hu et al., 2021); GRPO is a group-relative policy optimization method introduced in mathematical reasoning (Shao et al., 2024). We do not claim to reproduce their original numerical results. The proposed reward form is $r = \text{correctness} + 0.5$ agreement bonus, with correctness monitored independently. Under binary rewards, within-group advantage normalization can make an additive agreement component ineffective; diagnostics of reward and normalized advantage are therefore necessary.

Training uses $n \leq 128$, with larger-size and held-out-vocabulary evaluation kept separate. Latent instances must be disjoint across training and test. English context words such as *higher/lower* can overlap even when relation terms are held out, so predicate novelty is not full-context novelty. Fair comparison records example counts, actual training tokens, GPU time, steps, and trainable/update budgets; equal steps do not equalize compute. The local Apple M1/16-GiB preparation machine has no CUDA device; the training backend was validated on rented CUDA hardware by executed smoke-scale runs of all five arms under the declared dependency pins (Section 7). These runs establish pipeline feasibility and accounting only: no adapter has been performance-evaluated, and no adaptation improvement of any direction is claimed in this article.

6 Construction and Preprocessing Results

The following results concern actual dataset inventories and retained corpus/tokenizer measurements. They are not simulated model responses. Counts summarize the constructed collection; they are not random-sample estimates of a population accuracy, so model-performance confidence intervals and significance tests are inapplicable. The 33 split-by-size dataset records and the 236 native lexical publication records supply every numerical display in this section.

6.1 Dataset scale and the distinction between units

The collection contains 3,000 latent orders, 23,200 source queries, 5,637,600 prompt identities, and 7,819,200 comparison cells (Table 3). Source-order allocation is 450 training, 450 development, 300 pilot, and 1,800 test orders. The training allocation spans the frozen size grid, while optional training selects only $n \leq 128$; split membership does not imply selection for optimization. These counts describe different units: each source order supports multiple queries, and each query expands into 243 language/name/relation/presentation identities. Comparison cells additionally include matched contrasts and endpoint-to-interior comparisons. They are not 7.8 million independent reasoning trials.

Figure 2 shows why entity count, independent sample size, and workload must be kept separate. The full-test number of source orders is fixed per n, while the number of feasible distance conditions grows

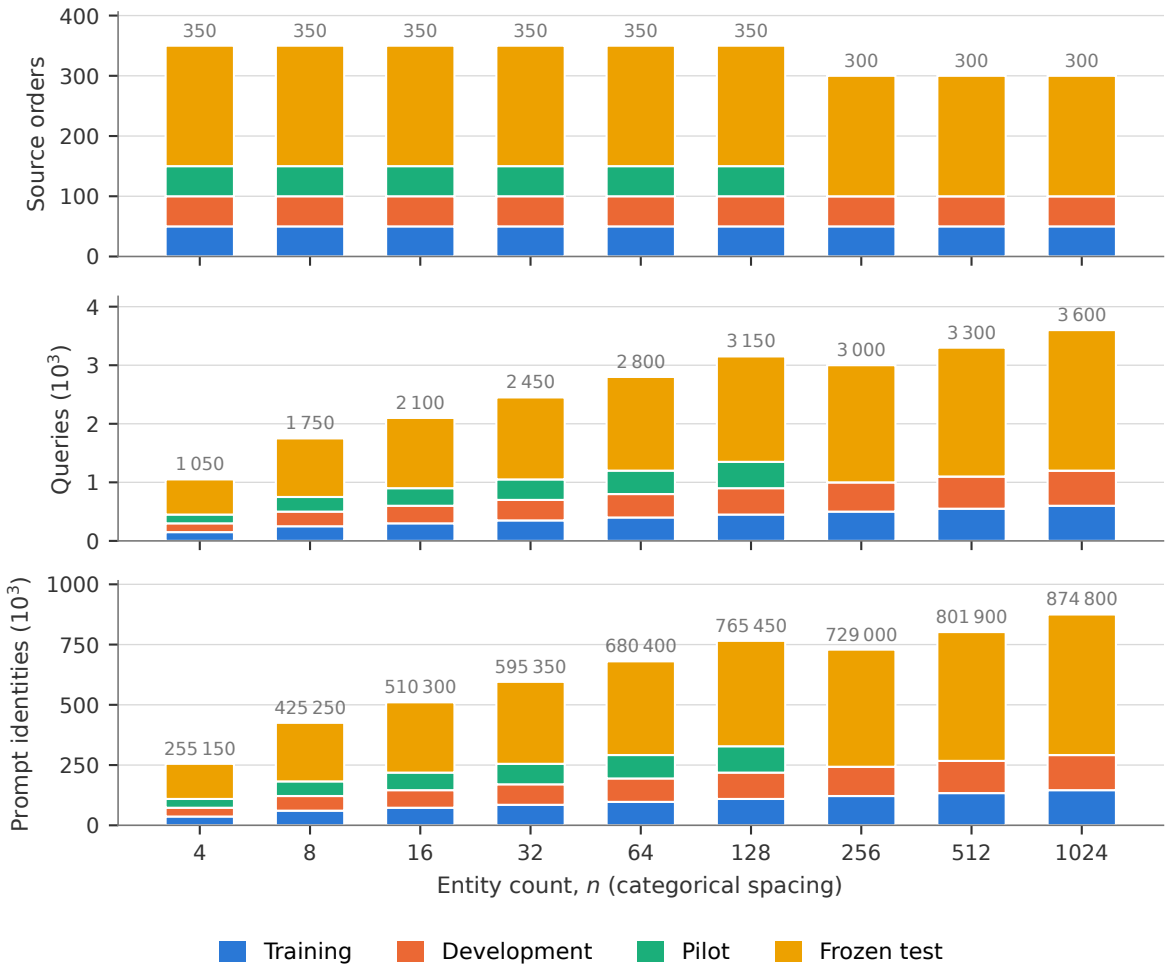


Figure 2: Verified construction counts by size and split. Source orders and derived queries/prompts are different units. This figure contains no model predictions.

with n. Pilot coverage stops at $n = 128$. The larger prompt workload at high n therefore reflects both presentation length and the design's query expansion; it is not evidence that a model performed more logical steps.

6.2 Independent corpus frequency and morphological realization

The retained lexical measurement uses the three corpus partitions of UD Turkish BOUN release r2.18, revision `0def94a229dcbc28d32192b0138b560c075c6d87`, under CC-BY-SA-4.0. BOUN's resource design and annotation context are described by Türk et al. (2022); the locked release, not the paper's headline counts alone, defines our measurements. Retained coverage includes 9,761 sentences, 101,052 eligible surface tokens, and 104,424 eligible lemma positions. The recorded normalization uses Unicode NFC and Turkish-aware case folding (I → ı, İ → i), with exact normalized n-gram matching. Surface n-gram opportunity counts differ with expression length; rates must retain their own denominators.

All 18 canonical inflected relation expressions have measured surface count zero, while their recorded lemma counts range from 10 to 390. Table 4 lists the paired forward/inverse lemma counts, and Figure 3 reports those values and directions. A measured zero is a property of this corpus and matching rule; it is not missing data and does not show that the expression is ungrammatical or absent from all Turkish. Conversely, a positive count of a lemma such as *üst* can include senses outside the intended ordering relation. Surface and lemma frequencies answer different questions.

For every zero surface count, an unsmoothed $\log_{10}$ frequency is undefined. Forward/inverse ratios

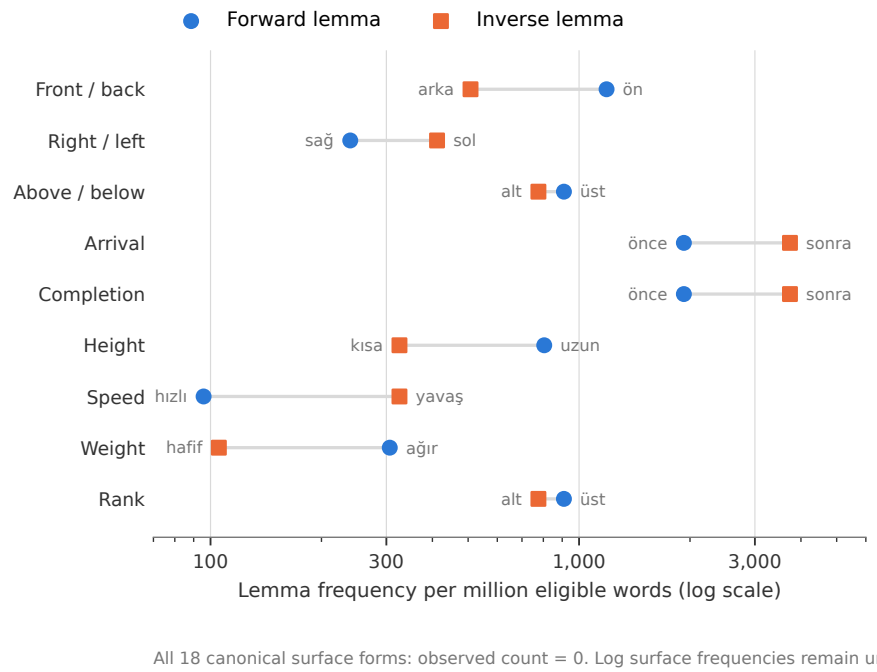


Figure 3: Observed canonical surface counts and lemma frequency proxies for the nine forward/inverse relation pairs. Surface zeros are measured zeros; lemma counts are not sense-disambiguated counts of the full relation expression. Lines connect paired forward and inverse proxies; they are not confidence intervals. No model outcome is represented.

with zero inverse denominators are likewise undefined; 0/0 is not set to one. Smoothing could define a different exploratory feature but would require an explicit new measurement specification. We retain the original zero/undefined distinction. These findings motivate careful lexical feature construction rather than a claim that lower frequency causes a reasoning error.

6.3 Tokenizer measurements

The isolated-vocabulary export has 216 records: six registered configurations, nine relations, two directions, and two languages. Counts cover the GPT-4o encoding configuration, the three Qwen configurations, and the two Llama configurations. The Llama tokenizer assets are license-gated upstream; they were vendored at the registered immutable revisions after license acceptance, with every file verified against its upstream Git object hash, rather than being inferred from another tokenizer (an earlier retained measurement records them as explicitly unavailable). A tokenizer measurement does not mean that model weights were loaded or that its answers were evaluated.

All 216 isolated-vocabulary records are measured (Table 5). Across the 18 matched relation-term pairs in each configuration, the median Turkish/English term-token ratio is 2.75 for the GPT-4o encoding, 2.25 for each Qwen configuration, and 2.00 for each Llama configuration; observed ratios across configurations range from 0.75 to 6. These are medians of matched ratios, not ratios of median token counts. They summarize the selected lexical inventory without inferential confidence intervals.

Figure 4 compares measured term tokenization across languages and directions on matched lexical entries. Shared-tokenizer configurations are shown as configurations, not treated as independent model families or independent replications. A ratio above one indicates a longer Turkish tokenization for that lexical measurement and tokenizer. It does not imply a corresponding accuracy loss. Isolated expressions, complete plain-text prompts, chat templates, and provider-billed usage have different scopes (Rust et al., 2021; Ahia et al., 2023; Petrov et al., 2023).

The corpus and tokenization audit provides features for RQ4; the executed pilot join (Section 7)

Table 4: Independent relation-lemma counts in UD Turkish BOUN r2.18 (104,424 eligible lemma tokens). These are lemma counts, not sense-disambiguated relation frequencies. All 18 canonical surface-form counts are zero. Shared lemmas recur across templates.

Relation	Forward lemma	Count	Inverse lemma	Count	f_F/f_I
Front / back	ön	124	arka	53	2.34
Right / left	sağ	25	sol	43	0.58
Above / below	üst	95	alt	81	1.17
Arrival	önce	201	sonra	390	0.52
Completion	önce	201	sonra	390	0.52
Height	uzun	84	kısa	34	2.47
Speed	hızlı	10	yavaş	34	0.29
Weight	ağır	32	hafif	11	2.91
Rank	üst	95	alt	81	1.17

Table 5: Tokenizer preprocessing coverage, not inference results. Each configuration has 18 Turkish and 18 English relation terms; ratios use matched isolated terms. The three Qwen configurations yield identical term counts, as do the two Llama configurations; neither family contributes independent tokenization samples.

Configuration	Measured	Median TR	Median EN	Median TR/EN
GPT-4o / o200k_base	36/36	4.0	2.0	2.75
Qwen2.5-0.5B / HF	36/36	5.0	2.5	2.25
Qwen2.5-1.5B / HF	36/36	5.0	2.5	2.25
Qwen2.5-1.5B / vLLM	36/36	5.0	2.5	2.25
Llama3.2-1B / HF	36/36	4.0	2.5	2.00
Llama3.2-3B / HF	36/36	4.0	2.5	2.00

yields direction-specific diagnostics but cannot identify the performance-association component from one relation family. The tokenized prompt preview is bounded to $n = 4$; the isolated-term export should not be described as an analysis of full chat length across the entire benchmark. Outcome analyses join by the exact latent/presentation coordinate and model/tokenizer revision.

6.4 What remains empirically unanswered

Table 6 prevents preprocessing observations from being mistaken for answers to RQ1–RQ5. The exploratory pilot evaluation of Section 7 supplies the first genuine accuracy-versus-n curves, $R/D/U$ decompositions, and declared-family tests on the pilot split and one relation family; the supplemental sweep of Section 7.7 extends them to all nine relation families and four model scales; and the completed human review of Section 7.8 backs the sampled stimuli's grammaticality and meaning preservation. Confirmatory status still requires the frozen-test design. The figures in the preceding subsections visualize actual construction/feature measurements and a conceptual pipeline. No source-paper model scores are relabeled as Turkish-CBE results.

7 Exploratory Model Evaluation and Linguistic Validation

This section reports the genuine model outcomes obtained under the framework and their linguistic validation, in chronological order: two complete registered pilot runs of 1,512 calls each, executed on 27 October 2026 on the pilot split (the first six subsections below); a supplemental quantized evaluation of four open-weight scales over 16,618 matched prompts each (Section 7.7); and the subsequently completed two-reviewer Turkish linguistic review of 360 stimuli sampled from the same frozen prompt population (Section 7.8). Every model result in this section is *exploratory*: the runs exercise the full execution and analysis machinery and motivate the confirmatory design, but the execution-gate receipts

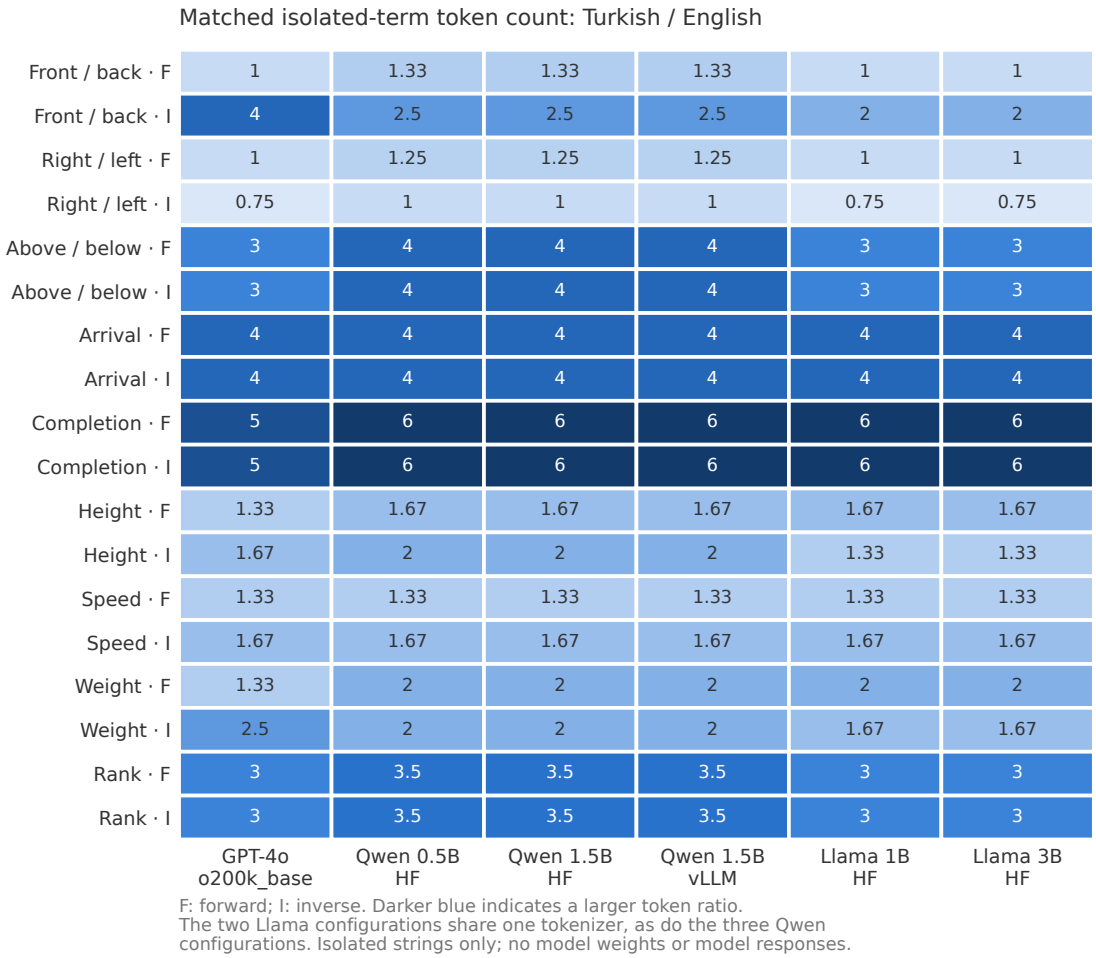


Figure 4: Verified isolated relation-term tokenization. Matched Turkish/English measurements and forward/inverse differences describe the lexical strings and retained tokenizer assets; these are not model-performance comparisons.

for confirmatory authorization were not issued, no frozen-test evaluation has been performed, and RQ1–RQ5 are not answered confirmatorily.

7.1 Runs, shared cohort, and execution integrity

Two registered model configurations were enabled by explicit user selection: Qwen2.5-0.5B-Instruct at a pinned immutable revision, executed locally with greedy decoding (CPU, float32), and the dated snapshot `gpt-4o-2024-08-06`, queried through its provider API at temperature zero. Both runs executed the same immutable plan cohort: entity counts $n \in \{4, 8, 16, 32\}$ with eight latent orders per size, endpoint, internal, and distance-controlled queries, the `spatial_front` relation, `native_shared` names, English canonical and equivalent conditions, and all seven Turkish conditions (canonical, general equivalent, and P1–P5). Each of the nine language/presentation conditions contains the same 168 planned queries; prompts are byte-identical across models.

Execution followed the audited protocol without exception: write-once plans re-derived and byte-compared at execution time, append-only raw request/response records with full lineage, and checkpointed resumption. One transient transport error occurred in the API run; following the ambiguity policy, the run halted and resumed only after an explicit, logged retry resolution acknowledging duplicate risk. Two earlier API run identifiers are retained as provenance of failed attempts (a local TLS misconfiguration before any billable call, and a provider spending-limit refusal with zero billed genera-

Table 6: Outcome availability. NA is unavailable, not zero performance. "Pilot (exploratory)" denotes the executed pilot-split runs of Section 7; no confirmatory frozen-test estimate is reported.

Target	Required evidence	Current status
Turkish–English gap	Matched genuine answers and all-planned accuracy	Pilot + sweep (exploratory)
Morphosyntactic robustness	Canonical/equivalent answers, $R/D/U$, coverage	Pilot + sweep (exploratory)
Complexity/distance profiles	Answers separated by n, d, and query type	Pilot + sweep (exploratory)
Lexical bias associations	Direction-specific predictions joined to measured features	Pilot (diagnostics only)
Stimulus linguistic validity	Two-reviewer sampled review with predeclared policy	Completed (sampled)
Reasoning/adaptation gains	Matched intervention runs and recorded budgets	Trained; evaluation gated

Table 7: Executed pilot runs. Both runs completed all planned calls with zero invalid and zero wrong-language answers under the strict parser.

Model	Execution	Calls	Invalid
Qwen2.5-0.5B-Instruct	local, greedy, CPU/float32	1 512/1 512	0
gpt-4o-2024-08-06	provider API, temperature 0	1 512/1 512	0

tions); neither contributes records to the analyzed run.

7.2 A degenerate responder: perfect invariance without competence

Qwen2.5-0.5B-Instruct emitted a constant answer per language on all 1,512 prompts: affirmative (*YES*) on every English prompt and negative (*HAYIR*) on every Turkish prompt, at every size, query type, and presentation. Because planned gold labels are balanced, marginal accuracy is exactly 0.500 in every stratum and the paired Turkish–English difference is zero everywhere. Raw invariance is perfect: $\hat{I} = 1$ within each language, with $\widehat{R} = \widehat{D} = 0.5$ and $\widehat{U} = 0$. The pooled answer rate is degenerate ($q \in \{0, 1\}$), so $I_e = 1$ and κ_{CBE} is undefined with the recorded reason `constant_pooled_answer`, exactly as Equation (11) prescribes.

This run is a concrete demonstration of the framework's central measurement argument rather than a mere null result. An evaluation reporting raw cross-presentation agreement would have scored this model as perfectly robust; the joint presentation of R, D, U, and the chance correction instead identifies a responder whose "consistency" consists entirely of shared correct-by-chance and shared wrong answers. No competence claim, positive or negative, beyond this behavioral characterization is made for the model.

7.3 GPT-4o: graded competence and a presentation cliff

All 1,512 GPT-4o answers were valid and in the requested language, so all-planned and valid-pair denominators coincide and valid-pair coverage is 1 in every stratum below. Figure 5 summarizes the marginal structure. Pooled all-planned accuracy is 72.0% on English ($N = 336$) and 57.8% on Turkish ($N = 1{,}176$). The two English conditions agree closely (canonical 71.4%, equivalent 72.6%), while the Turkish conditions separate: canonical 67.9% and P1 68.5% at the top; equivalent 57.7%, P2 58.3%, P4 61.9%, and P5 53.6% in between; and P3, the alternative constituent construction, at 36.9% — below the 50% balanced baseline.

Accuracy declines monotonically with entity count in both languages (English 100% → 58.9% and Turkish 77.4% → 51.5% from $n = 4$ to $n = 32$) and with latent query distance within the distance-controlled stratum (70.5% at $d = 1$ to 36.1% at $d = 16$). The pooled query-type ordering is endpoint 76.4% > internal 64.9% > distance-controlled 55.0%. Sustained below-baseline accuracy at large d indicates a systematic directional error rather than random guessing. Predicted answer rates also differ

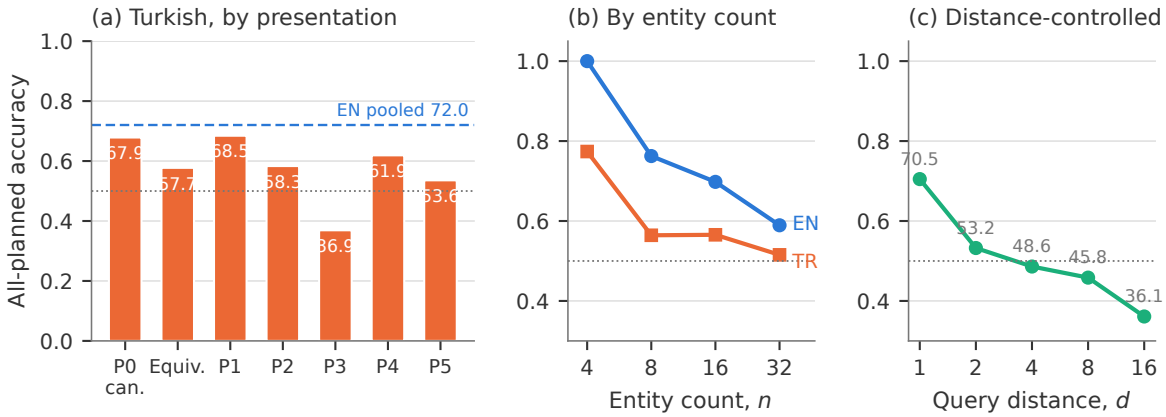


Figure 5: Exploratory pilot accuracy of `gpt-4o-2024-08-06` (all answers valid; all-planned accuracy; dotted line: 50% balanced baseline). (a) Turkish accuracy by presentation condition with the pooled English reference; 168 prompts per condition. (b) Accuracy by entity count and language. (c) Pooled accuracy of distance-controlled queries by latent query distance d; per-distance counts vary by design (288, 216, 216, 144, and 72 prompts). Pilot split, one relation family, stimulus review in Section 7.8; these curves are not confirmatory answers to RQ1–RQ3.

Table 8: Pooled descriptive paired profiles for GPT-4o (168 jointly valid pairs per row; valid-pair coverage 1.0, so Equations (9)–(10) coincide with their all-planned counterparts). q is the pooled affirmative rate of Equation (11). Pooling crosses size and query-type strata, so these values are summaries of the executed cohort, not stratum estimands; per-stratum profiles with cluster-bootstrap intervals are in the released analysis tables.

Comparison	R	D	U	q	κ_{CBE}
TR canonical → TR equivalent	0.458	0.202	0.339	0.652	0.253
TR canonical → TR P1	0.565	0.202	0.232	0.640	0.496
TR canonical → TR P2	0.500	0.238	0.262	0.667	0.411
TR canonical → TR P3	0.298	0.250	0.452	0.673	−0.027
TR canonical → TR P4	0.542	0.244	0.214	0.631	0.540
TR canonical → TR P5	0.446	0.232	0.321	0.667	0.277
EN canonical → TR canonical	0.565	0.173	0.262	0.560	0.469
EN equivalent → TR equivalent	0.476	0.173	0.351	0.568	0.284

by language: the affirmative rate is 0.476 on English but 0.680 on Turkish against balanced gold labels, i.e. Turkish errors skew toward *EVET*.

Table 8 gives the pooled paired decomposition. The P3 contrast is qualitatively different from every other Turkish perturbation: robust correctness falls to $\widehat{R} = 0.298$, presentation dependence rises to $\widehat{U} = 0.452$, and the pooled chance-corrected agreement is approximately zero ($\kappa_{\mathrm{CBE}} = -0.027$) — the model's answers under the canonical and P3 realizations of the same latent query are statistically unrelated. The cross-language rows show the second mechanism the decomposition separates: moving from English to Turkish converts correct answers into presentation-dependent ones (U up to 0.351) rather than into shared errors (D stable at 0.173).

7.4 Inferential results at pilot power

Declared-family inference behaves as the design anticipates at this scale. Across the 70 matched Turkish–English strata, the accuracy difference is negative in 31 and positive in 10 (mean −0.098), but no individual eight-pair stratum survives Holm correction: the direction is consistent while per-stratum McNemar tests are underpowered, which is precisely the regime the frozen-test sample sizes were chosen to escape. The P3 effect, by contrast, is strong enough to register even here: the canonical-versus-P3 family contains the only Holm-surviving strata in the study (two strata at $n = 32$, endpoint and distance-controlled,

Table 9: Direction-specific pilot diagnostics for GPT-4o (all planned gold labels balanced at 0.500; all answers valid). Forward queries use *önündedir* / *in front of*; inverse queries use *arkasındadır* / *behind*. These are descriptive shares on one relation family; direction changes the lexical predicate, the construction, and tokenization together, so no single cause is identified.

Language	Query direction	*N*	Accuracy	Affirmative share
Turkish	forward	588	0.639	0.568
Turkish	inverse	588	0.517	0.793
English	forward	168	0.768	0.399
English	inverse	168	0.673	0.554

each with difference −0.75 and Holm-adjusted $p = 0.031$). Cluster-bootstrap intervals accompany every reported stratum in the released tables; pooled values above are descriptive.

7.5 Exploratory lexical-direction diagnostics (RQ4)

The retained lexical pipeline was additionally executed joined to both completed runs, producing the declared answer-share diagnostics, lexical-direction profiles, Spearman associations, and controlled regression specifications on the pilot observations. Its most interpretable output is a pronounced *direction asymmetry* in GPT-4o (Table 9): against balanced gold labels, queries asked with the inverse predicate (Turkish *arkasındadır*, English *behind*) are answered affirmatively far more often and less accurately than forward queries (*önündedir* / *in front of*). The asymmetry appears in both languages and is largest in Turkish, where the inverse-direction affirmative share reaches 0.793; it accounts for most of the pooled Turkish affirmative bias reported above. The predeclared response-tendency diagnostic flags Qwen2.5-0.5B in every stratum (excessive *no* in Turkish, excessive *yes* in English), while GPT-4o pools within thresholds overall and crosses the affirmative threshold only in inverse-direction Turkish strata.

This asymmetry aligns suggestively with the preprocessing measurements: the forward lemma *ön* is more than twice as frequent as the inverse lemma *arka* in the retained corpus measurement (Table 4), and for the GPT-4o encoding the inverse Turkish predicate is four times its English counterpart's token length while the forward pair is matched (Figure 4). The design, however, cannot promote this alignment to an association: the pilot cohort contains a single relation family, so each correlation cell holds at most two lexical units, and the retained Spearman and regression outputs record exactly that — the large majority of cells return `constant_input`, `insufficient_pairs`, `insufficient_rows`, or `rank_deficient` rather than a coefficient, as the protocol requires of unsupported or collinear fits. RQ4's association component therefore remains unanswered; answering it requires the multi-relation frozen design, where the nine relation families provide the cross-vocabulary variation this cohort lacks.

7.6 Training backend execution (RQ5)

The optional adaptation backend was executed for the first time on rented CUDA hardware (RTX 6000 Ada, 48 GB, bf16) under the configuration's declared dependency pins, which this run validates. All five training arms of the pilot profile — LoRA SFT, paired SFT, SFT with the JS consistency term, and both GRPO objectives — completed on Qwen2.5-1.5B-Instruct from a hash-verified local weights bundle, with immutable plans, zero gate blockers, recorded hardware and token budgets, and deterministic seeding. The pilot profile is smoke-scale by design (two optimizer steps per arm), so these runs validate the training machinery and its accounting, not any adaptation effect. Both GRPO arms recorded one zero-gradient step, the exact diagnostic the plan's warnings predict for binary prompt-normalized advantages. The matched base-versus-adapter evaluation remains unexecuted: the protocol's evaluation design is frozen at nine sizes on the test split (10,224 calls per arm), which requires a dataset-bound execution-gate review receipt that has not been issued (the completed stimulus review of Section 7.8 is a separate instrument); no reduced evaluation is expressible within the validated configuration, deliber-

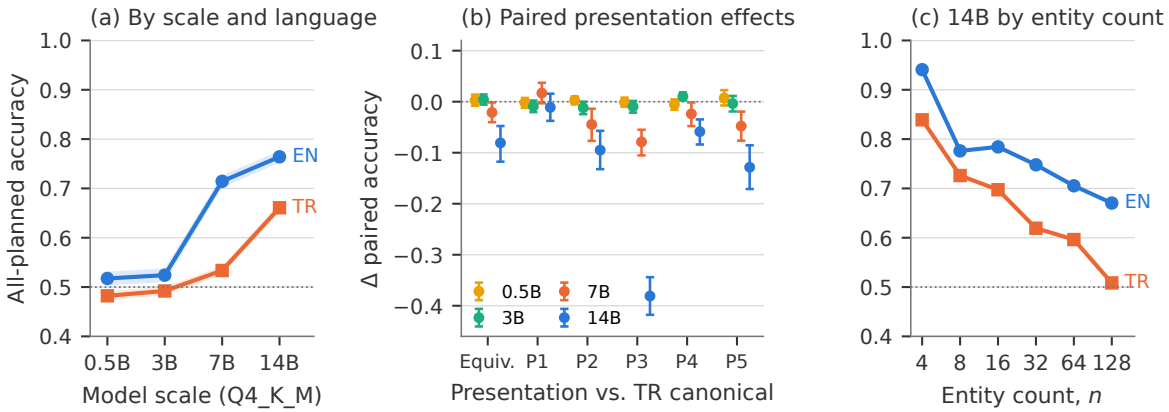


Figure 6: Supplemental quantized local evaluation of the Qwen2.5 instruct family (Q4_K_M; all-planned accuracy over 16,618 prompts per model; dotted lines: 50% balanced baseline and zero effect). (a) Accuracy by model scale and language with Wilson bands. (b) Query-matched paired accuracy change of each Turkish presentation against the Turkish canonical form, with latent-cluster bootstrap 95% intervals; pooled single-condition accuracies are compositionally confounded and are not shown. (c) Accuracy by entity count for the 14B model. Pilot split, quantized supplemental arm, stimulus review in Section 7.8; not confirmatory answers to RQ1–RQ3.

ately. RQ5 therefore remains empirically unanswered, with trained adapters retained should that receipt be produced.

7.7 Supplemental quantized local evaluation: an open-weights scale sweep

To test whether the pilot's findings are idiosyncrasies of one proprietary model, the frozen protocol was additionally executed across the Qwen2.5 instruction-tuned family at four scales (0.5B, 3B, 7B, 14B) in a *supplemental quantized local arm*: Q4_K_M GGUF weights served by Ollama (weight digests recorded per run), greedy decoding at temperature 0 with fixed seed, the audited message constructor with runtime prompt-hash re-verification, and the registered strict parser. Because quantized weights are not the registered fp/bf16 snapshots, this arm is labeled on every record and is never pooled with registered-provider results; agreement across arms is read as qualitative replication, not equivalence. The expanded manifest holds 16,618 prompt identities drawn exclusively from the pilot split (all nine relation families, sizes $4 \leq n \leq 128$, balanced paired sampling across languages, presentations, and naming controls; the frozen test sizes remain untouched), and every prompt passed an automated semantic validation of its stated facts and gold label before execution. All four arms answered all 16,618 prompts with zero transport failures; unparseable responses (0, 25, 184, and 247 from 0.5B to 14B, the latter coinciding with output-length truncation) are retained and scored as errors under the all-planned estimand. Like the pilot, this expanded run exceeds the predeclared human-review threshold; the stimulus review of Section 7.8 was subsequently completed on this run's prompt population, but no execution-gate receipt was issued, and the arm remains exploratory (quantized weights, pilot split).

The sweep reproduces both pilot regimes and locates the transition between them (Figure 6a). The 0.5B and 3B models are near-constant responders — 0.5B answers *hayır* on 95.8% of Turkish and *yes* on 99.8% of English calls, and its paired English–Turkish profile is the degenerate signature in pure form ($R = 0.017$, $U = 0.970$: on almost every matched pair exactly one language is "correct" because the two constant answers disagree). Discrimination emerges at 7B in English only (71.4% vs. 53.4%; paired gap −0.176, cluster-bootstrap 95% CI [−0.241, −0.106]) and reaches both languages at 14B (76.4% vs. 66.1%; paired gap −0.075, CI [−0.126, −0.020]). Because pooled pairs share latent orders, ordinary McNemar inference is undefined for these contrasts; all language and presentation intervals are latent-cluster bootstrap contrasts and are labeled exploratory.

Two pilot findings replicate in open weights and sharpen with scale. First, the P3 presentation cliff: query-matched P3 costs the 14B model 38.1 points of paired accuracy against the Turkish canonical

form (CI [−0.418, −0.344]; 554 vs. 40 discordant pairs), dropping P3 accuracy to 43.1% — below the balanced baseline — while the same contrast at 7B is already credible at −7.9 points and absent in the constant responders (Figure 6b). Per-stratum agreement decays with complexity (κ_{CBE} for canonical-versus-equivalent at 14B: 0.57 at n=4 to 0.12 at n=128), and for canonical-versus-P3 it turns negative at n=16 and n=32: systematic answer flips, not noise. Second, the complexity decay: 14B Turkish accuracy falls monotonically from 83.9% at n=4 to 50.8% — chance — at n=128 (English 94.1% to 67.0%; Figure 6c), mirroring the GPT-4o pilot curve. The naming family, whose pairs are cluster-unique and therefore admit exact inference, adds one Holm-surviving result: at 14B, replacing native Turkish entity names with shared symbolic labels costs 11.7 points (McNemar b=59, c=24, Holm-adjusted $p = 3.8 \times 10^{-4}$), while English transliteration of the same names costs nothing measurable — once competence exists, entity-name surface form matters, and the deficit is not explained by transliteration alone.

7.8 Completed two-reviewer Turkish linguistic review

The human review that the construction audit required was executed by two independent Turkish-proficient reviewers on a 360-item stratified sample drawn from the frozen pilot-split prompts (all nine relation families, every Turkish presentation, P3 deliberately oversampled at 144 items; annotation protocol, blinding, and the inclusion policy were fixed before any judgment was entered). Each reviewer independently rated grammaticality and naturalness (1–5), semantic equivalence of each transformed prompt against its canonical counterpart, and clause-level ambiguity. The predeclared policy admits a transformed item to confirmatory analysis only when both reviewers rate grammaticality ≥ 4 and agree the transformation is meaning-preserving.

The outcome is sharply structured. Semantic equivalence is unanimous: both reviewers judged all 324 transformed items meaning-preserving, and no item was flagged ambiguous. Grammaticality excluded exactly one construction: the P3 realization of *temporal_completion*, whose 16 sampled items one reviewer rated ungrammatical (grammaticality 2, "devrik cümle yapısı"), leaving 344 of 360 items confirmatory-eligible. Naturalness — which the policy deliberately records but does not gate on — is the one contested dimension: exact inter-reviewer agreement is low (33%, Cohen's $\kappa = 0.06$), the two reviewers' P3 naturalness medians are 4 versus 2, and only 6 of 144 P3 items received ≥ 4 from both. (Chance-corrected agreement is degenerate for grammaticality and ambiguity because one or both reviewers used a single category almost everywhere; raw agreement is 96% and 100%.)

The review therefore narrows, but does not fully close, the stimulus-artifact explanation of the P3 cliff. A review-driven sensitivity reanalysis removes the excluded relation entirely from the P3 contrast: the 14B query-matched deficit survives unchanged or deepens (−0.381 over all nine relations versus −0.408, cluster-bootstrap 95% CI [−0.446, −0.370], with *temporal_completion* removed; the 7B contrast is unchanged at −0.080). The P3 collapse is thus measured on stimuli that two independent reviewers unanimously certify as meaning-preserving and (outside one excluded construction) grammatical, while their discourse naturalness remains contested — the honest residual reading is that models fail on grammatical, truth-preserving but discourse-atypical constituent orders, a regime in which a competent human reader does not lose the ability to evaluate the stated facts. Review artifacts, per-item decisions, and agreement statistics are released with the framework (`validation/review_results/`); the separate execution-gate receipts for runs above the declared call threshold were not issued, so no evaluation in this manuscript claims confirmatory status.

7.9 What this evaluation does and does not establish

These experiments establish that the end-to-end protocol — immutable planning, gated execution, strict parsing, paired analysis with declared denominators, and multiplicity control — runs unmodified on local, provider-hosted, and quantized local models, and that the paired decomposition discriminates in practice between degenerate constant responders and models with graded, presentation-sensitive competence, with the same qualitative structure recurring from the registered pilot to the digest-pinned nine-relation sweep. They do not establish a confirmed Turkish–English gap, a morphology effect, or a com-

Table 10: Consolidated exploratory findings. "Sweep" denotes the supplemental quantized Qwen2.5 evaluation (16,618 matched prompts per model, pilot split, all nine relation families); paired effects are query-matched with latent-cluster bootstrap 95% intervals. pp = percentage points.

Dimension	Main observation	Supporting evidence	Interpretation
Model scale	All-planned accuracy rises from chance to 69.2% across 0.5B–14B	Sweep: 49.3%, 50.2%, 58.7%, 69.2% (Fig. 6a)	Scale-associated discrimination in the evaluated family
Language	English accuracy exceeds Turkish once models discriminate	14B: 76.4% vs. 66.1%; paired gap −7.5 pp [−12.6, −2.0]; 7B: −17.6 pp [−24.1, −10.6]	Language-dependent behavior on matched problems
Presentation	P3 constituent order produces a large accuracy decrease	14B: −38.1 pp [−41.8, −34.4], discordant 554 vs. 40; −40.8 pp [−44.6, −37.0] after the review exclusion	Constituent-order sensitivity on meaning-preserving input
Complexity	Accuracy decreases monotonically with entity count	14B Turkish: 83.9% at n=4 to 50.8% at n=128 (Fig. 6c)	Input-size sensitivity
Naming	Shared symbolic labels reduce accuracy at 14B	−11.7 pp, McNemar b=59, c=24, Holm $p = 3.8 \times 10^{-4}$	Entity-representation sensitivity
Invariance	Constant responders show perfect agreement without competence	0.5B: $R = 0.017$, $U = 0.970$ on EN–TR pairs; κ_{CBE} undefined	Stability is not competence

plexity law: everything remains on the pilot split; strata share latent orders; and the supplemental arm is quantized and therefore not precision-comparable to registered snapshots. Section 8 synthesizes the findings and their interpretive boundaries, including what the completed review does and does not resolve about P3. These results size and motivate, but do not replace, the preregistered frozen-test evaluation.

8 Discussion

Table 10 consolidates the main empirical findings of Section 7; the subsections below synthesize what they establish and where their interpretive boundaries lie. All findings are exploratory and describe the evaluated models, cohorts, and settings.

8.1 Correctness and presentation invariance are distinct

The paired decomposition supports questions that a marginal accuracy table cannot answer. A small average difference between canonical and equivalent prompts can coexist with large, offsetting correct-to-wrong and wrong-to-correct flows; conversely, agreement can improve because a model repeats a wrong label. Equations (9)–(10) specify the evidence needed to distinguish these cases: shared-valid $R/D/U$, all-planned accuracy and strict R, invalid rates, and coverage. The evaluated models realize both failure modes. Qwen2.5-0.5B attains perfect raw invariance by answering near-constantly per language (Turkish *hayır* on 95.8% of Turkish calls, *yes* on 99.8% of English calls), and its English–Turkish pairs show the degenerate signature in pure form: $R = 0.017$ with $U = 0.970$, because the two constant answers disagree and exactly one side is scored correct on almost every pair. Any protocol that reported agreement without the decomposition would have certified this model. The chance correction has a correspondingly limited role: constant outputs make κ_{CBE} undefined by construction, and its pooled-marginal baseline discounts one particular independent-answer bias reference without modeling failure causally or proving stochastic independence. Correctness and invariance remain behavioral properties;

explanation text adds an observation channel, not access to the model's internal computation (Turpin et al., 2023).

8.2 Turkish constituent-order sensitivity

The P3 result is the sharpest finding. On query-matched pairs, the alternative constituent construction costs the 14B model 38.1 points of paired accuracy (95% CI [−41.8, −34.4] pp; 554 vs. 40 discordant pairs), taking Turkish P3 accuracy below the balanced baseline, with a credible −7.9 pp precursor already at 7B and no effect in the constant responders; the registered pilot found the same collapse in GPT-4o (36.9% under P3). The completed review (Section 7.8) bounds its interpretation from both sides. On one side, both reviewers unanimously judged the sampled transformations meaning-preserving, and the one construction that failed grammaticality for one reviewer (*temporal_completion* under P3) is removed by a sensitivity reanalysis in which the effect persists and deepens (−40.8 pp, [−44.6, −37.0]). A pure stimulus-grammaticality artifact is therefore ruled out for the retained constructions. On the other side, the reviewers' naturalness judgments genuinely differ (P3 medians 4 vs. 2), so the transformations cannot be claimed uniformly natural in every discourse context; this documented linguistic variation qualifies, but does not invalidate, the finding. The supported conclusion is that the evaluated models exhibit substantial behavioral sensitivity to meaning-preserving Turkish constituent-order variation, including constructions both human reviewers accepted as grammatical. The effect does not by itself identify *which* capacity fails—logical inference, syntactic parsing, or the representation of discourse-atypical input—only that a competent human reader of the same sentences does not lose the ability to evaluate the stated facts.

8.3 Model scale and multilingual reasoning

The evaluated Qwen2.5 family progresses from strongly biased, near-constant response behavior (0.5B, 3B) toward discriminative performance, in English before Turkish (7B: 71.4% vs. 53.4%) and then in both (14B: 76.4% vs. 66.1%). Scale is associated with higher accuracy, a narrower but persistent language gap, and—critically—*increasing* absolute presentation sensitivity: the P3 deficit grows from nothing (constant responders) through −7.9 pp (7B) to −38.1 pp (14B), and per-stratum κ_{CBE} for canonical-versus-equivalent decays with complexity at 14B (0.57 at n=4 to 0.12 at n=128). Improved canonical accuracy therefore does not guarantee equivalent robustness across alternative linguistic presentations: the most capable evaluated model is simultaneously the most presentation-sensitive in absolute terms. We do not claim that 7B marks a universal emergence threshold, that parameter count is the sole causal factor, or that the pattern extends beyond this family; the supplemental experiments use quantized weights from a single model lineage, and response bias, instruction tuning, and training data co-vary with scale.

8.4 Complexity and entity representation

The endpoint shortcut makes a strong endpoint score ambiguous as evidence of transitive reasoning; internal and distance-controlled questions reduce that specific shortcut without eliminating every efficient strategy. Within those boundaries, the exploratory profiles are consistent: accuracy declines monotonically with entity count in every discriminating model (14B Turkish reaches chance at n=128), the pilot's endpoint > internal > distance-controlled ordering matches the shortcut analysis, and below-baseline accuracy at large query distance indicates a systematic directional error rather than noise. Changes with n may still reflect retrieval, distraction, context length, or the distribution of queried distances. Entity representation matters independently: replacing native Turkish names with shared symbolic labels costs the 14B model 11.7 points (the only Holm-surviving McNemar result in the sweep), while English transliteration of the same names costs nothing measurable—so the deficit is not explained by transliteration alone. The pilot's direction asymmetry (Table 9) aligns suggestively with measured frequency and tokenization asymmetries, but the single-relation cohort leaves every lexical association formally unidentified, and the retained specifications record refusals rather than coefficients, exactly as the protocol requires.

8.5 Implications for multilingual benchmark design

These findings argue that multilingual reasoning benchmarks should report matched correctness and presentation stability rather than accuracy alone. An accuracy-only leaderboard would have ranked the constant responders near chance—correctly—but would also have certified their perfect cross-presentation agreement had invariance been reported without correctness, and it would have missed that the strongest evaluated model loses more than a third of its paired accuracy under one meaning-preserving Turkish construction. Matching latent instances across languages, holding answers fixed under transformation, declaring denominators, and separating R, D, and U are each individually simple; their combination is what makes the failure modes visible. The same structure also disciplines interpretation: measured zeros in corpus frequency, tokenization ratios, and naturalness disagreement are reported as measured properties with explicit scopes, not converted into causal explanations (Rust et al., 2021; Ahia et al., 2023; Petrov et al., 2023).

8.6 Scope of the empirical evidence

All model results are exploratory: pilot-split cohorts, one quantized model family in the sweep, no issued execution-gate receipts, and no frozen-test evaluation. The framework's inferential machinery was sized for the frozen test, and the pilot vindicates that sizing in both directions—eight-pair strata detect nothing for the consistently negative Turkish–English contrast, while the P3 effect survives Holm correction even at pilot scale. Null, adverse, or non-replicating confirmatory outcomes remain possible and would be incorporated without changing definitions after observing results; Section 9 records the limitations and Section 9.2 the confirmatory agenda.

9 Limitations and Future Work

9.1 Limitations of the present study

Exploratory scope. All model evidence is exploratory and comes from the pilot split: the two registered pilot runs (one relation family, one name control, $n \leq 32$) and the supplemental sweep (nine relation families, $n \leq 128$) with quantized Q4_K_M weights that are not precision-comparable to registered fp/bf16 snapshots. The sweep covers a single model lineage (Qwen2.5 instruct), so model-family diversity is limited; GPT-4o adds one proprietary configuration on the narrower pilot cohort. No confirmatory frozen-test estimate exists, and no adaptation improvement has been evaluated—the executed training runs are smoke-scale backend validation only.

Linguistic coverage and validation. The renderer covers a closed grammar and a finite relation and presentation inventory; Turkish morphology and word order are richer than the implemented templates, and P4/P5 do not identify a pure morphology manipulation. Shared generation/parsing code can conceal a common-mode mistake. The completed human review (Section 7.8) certifies semantic equivalence unanimously and grammaticality outside one excluded construction, but it is sampled rather than exhaustive, and its naturalness dimension shows low inter-reviewer agreement; it is not a proof that every transformation is natural in every discourse context. Minimal-pair results in prior literature cannot validate our stimuli automatically.

Task and dependence structure. Strict total orders exclude partial orders, contradictory premises, ties, and open-world uncertainty. Renamed orders are structurally isomorphic. Multiple queries, names, relations, and presentations share each latent source, so prompt counts overstate independent sample sizes by orders of magnitude; all inference in this article clusters on latent orders, and exact split separation does not establish absence of pretraining contamination.

Frequency and tokenization proxies. BOUN has finite genre and sense coverage; surface n-gram and lemma opportunities differ; exact canonical zeros need not reflect general usage; and lemma counts are not relation-sense counts. Tokenizer measurements cover six registered configurations, with shared-tokenizer configurations not independent. Isolated-string measurements exclude chat overhead and billed usage. With one relation family in the pilot join, no lexical association is identified.

Inference and missingness. Context overflow, wrong-language answers, truncation, and provider nondeterminism may be systematically associated with difficult items; conditional CBE statistics therefore always carry coverage and all-planned counterparts. Cluster-bootstrap intervals depend on sufficient independent latent orders, and the declared Holm families do not provide study-wide simultaneous error control. Where pooled pairs share latent clusters, ordinary McNemar inference is reported as undefined and replaced by exploratory cluster-bootstrap contrasts.

9.2 Future work

The confirmatory agenda is specified and frozen but unexecuted. It comprises: (1) the preregistered frozen-test evaluation under the declared primary families (test split, sizes up to $n = 1024$); (2) the formal execution-gate approvals—dataset-bound two-reviewer receipts and pilot approval—that authorize runs above the declared call threshold, which are future methodological procedures rather than requirements of the present exploratory study; (3) additional model-family comparisons beyond Qwen2.5, including full-precision registered snapshots; (4) the matched base-versus-adapter evaluation of the five already-trained arms under the frozen 10,224-call-per-arm design; (5) reasoning-prompting interventions under the declared budget and faithfulness caveats; (6) expanded linguistic and relation-family coverage, including naturalness-controlled constituent-order stimuli that could close the residual P3 ambiguity; and (7) multi-relation lexical-familiarity and tokenization analyses, which become identifiable once the frozen design supplies cross-vocabulary variation.

10 Conclusion

Turkish-CBE contributes, first, a controlled bilingual reasoning benchmark: a deterministic generator whose latent total orders, exact answers, matched Turkish–English realizations, controlled presentation transformations, and split-separated instances allow the same question to be asked across languages and grammatical forms. Second, it contributes a correctness-aware robustness evaluation: the denominator-aware decomposition of paired outcomes into robust correctness, shared error, and presentation dependence, reported with coverage and a chance-corrected agreement reference. The two are designed together because neither accuracy nor invariance alone characterizes reasoning reliability—a point the evaluated models demonstrate concretely, with small models achieving perfect cross-presentation agreement by answering constantly and contributing no evidence of competence.

Third, the completed exploratory evaluation characterizes linguistic presentation sensitivity in real systems. Across the evaluated Qwen2.5 scales, discrimination emerges in English before Turkish, and the strongest model remains the most presentation-sensitive in absolute terms: a truth-preserving Turkish constituent-order alternative costs it 38.1 points of query-matched paired accuracy, an effect that persists at 40.8 points after excluding the one construction the completed two-reviewer linguistic review rejected on grammaticality. Accuracy also decays with entity count to chance at the largest evaluated sizes, and symbolic entity labels impose a measurable cost once competence exists. For multilingual evaluation practice, these findings indicate that benchmarks should report matched correctness and presentation stability jointly, with declared denominators, rather than accuracy on a single realization.

All model results are exploratory—pilot split, one quantized model family in the sweep, naturalness contested in the human review—and the preregistered frozen-test evaluation, additional model families, and the matched adapter contrast remain future work (Section 9.2). The scientific takeaway stands independently of those extensions: reasoning reliability cannot be adequately characterized by accuracy

on a single linguistic presentation; it also requires measuring whether correct conclusions remain stable across meaning-preserving alternatives.

Author Contributions

Conceptualization, methodology, software, validation, formal analysis, data curation, and writing — original draft: both authors. Supervision and writing — review and editing: M.O. and O.B. All authors have read and agreed to the published version of the manuscript.

Data and Code Availability

The complete framework — the frozen grid construction, renderers, audited analysis and statistics modules, execution plans, run records of every executed evaluation (including failed attempts retained as provenance), review instruments with both reviewers' completed annotations, and the scripts that regenerate every table and figure in this manuscript — is available at https://github.com/okanss/turkish-cbe (public upon publication). The frozen dataset manifest and all result files are hash-bound as described in Appendix C. License-gated tokenizer assets are not redistributed; the repository documents how to obtain them.

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Conflicts of Interest

The authors declare no conflict of interest.

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A Elementary Identities and Chance References

Proposition 1. *For a binary task with a common gold answer and two jointly valid answers, the three outcome masses partition the sample, and* $A_{\text{valid}} = R + U/2$, $I = R + D$.

Proof. Each pair is both correct, both wrong, or exactly one correct, so $R + D + U = 1$. The two-presentation correctness count is two for a both-correct pair, zero for a both-wrong pair, and one for a discordant pair. Dividing by two yields $A_{\text{valid}} = R + U/2$. In a binary task, both-wrong answers are the same wrong label; agreement is therefore $R + D$. The argument requires the same valid-pair support and the same gold proposition. □

Proposition 2. *For* $N > 0$ *and* $M > 0$, *the strict robustness identity is* $R_{\text{strict}} = vR$, *and the accuracy-constrained bounds are* $\max(0,\ 2A_{\text{valid}} - 1) \le R \le A_{\text{valid}}$.

Proof. Both robustness numerators count the same both-correct valid pairs; their denominators are M and N. Multiplying by M/N gives the first identity. Since $U = 2(A_{\text{valid}} - R) \ge 0$, $R \le A_{\text{valid}}$. Since $D = 1 - R - U = 1 - 2A_{\text{valid}} + R \ge 0$, $R \ge 2A_{\text{valid}} - 1$; probabilities are also nonnegative. □

Proposition 3. *Let* p_0, p_1 *be the two YES marginals on jointly valid pairs and* $q = (p_0 + p_1)/2$. *The CBE pooled reference and Cohen's separate-marginal reference satisfy*

$$I_{e,\text{CBE}} - I_{e,\text{Cohen}} = \frac{(p_0 - p_1)^2}{2}, \tag{14}$$

where $I_{e,\text{Cohen}} = p_0 p_1 + (1 - p_0)(1 - p_1)$.

Proof. Expanding the pooled expression gives $1 - p_0 - p_1 + (p_0 + p_1)^2/2$. Subtracting $1 - p_0 - p_1 + 2p_0p_1$ yields the stated square. Thus the references agree when the marginals agree; otherwise they are distinct. For fixed observed agreement below one, increasing the chance reference decreases the coefficient wherever its denominator is positive. This is an algebraic property, not an empirical model result or a new estimate of reliability. □

B Illustrative Turkish Equivalence and Its Boundary

The following forms illustrate the closed-grammar contract; they are not model outputs:

TR forward:	Ayşe, Bora'nın önündedir.
TR converse:	Bora, Ayşe'nin arkasındadır.
EN forward:	Ayşe is in front of Bora.
EN converse:	Bora is behind Ayşe.

The equivalence presupposes the specified linear interpretation and the same entities. Proper-name suffixes use the pronunciation and vowel/consonant environment of the supplied name; ASCII transliteration is a separate name control, not an alternative spelling silently substituted in a Turkish condition.

For distinct a, b in a strict total order, trichotomy gives $\neg(a \prec b) \Leftrightarrow b \prec a$. This justifies a formal negation transformation only under those assumptions. It does not license unrestricted natural-language negation, ties, or partial-order interpretations. Optional P6 stimuli require their own grammar and human review and are excluded from the frozen core described in the main text.

C Data, Figure, and Bibliography Provenance

Every new quantitative figure reads the retained dataset-statistics CSVs, lexical-publication CSVs, or normalized pilot-run records. The plot manifest records source SHA-256 hashes, deterministic derived values, and generated asset hashes. Surface count zero is retained as zero, and undefined logarithms/ratios remain undefined. The retained plain-text exports are not reclassified as provider-billed usage. Figure 1 is a conceptual diagram with an illustrative instance; Figures 2–4 show construction/preprocessing measurements only; Figure 5 shows exploratory pilot model outcomes read from the append-only normalized records of the runs in Table 7.

Three lexical measurement snapshots are retained. The baseline snapshot (`results/lexical/baseline`) predates the Llama tokenizer assets and records their 72 records as explicitly unavailable. The preprocessing snapshot (`results/lexical/with_llama`), which the manuscript figures read, was produced after the license-gated `meta-llama` tokenizer files (and each configuration's public `config.json`) were vendored into the local asset store at the registered immutable revisions, with every file verified against its upstream Git object hash before any offline load; all 144 previously measured term counts are unchanged between the snapshots, and no model weights were added. The asset-verification policy was widened accordingly (tokenization measurement version 0.7.0 → 0.8.0): a bundle must contain the fast-tokenizer pair and a license file, while the split BPE vocabulary files remain verified where a repository ships them. The joined snapshot (`results/lexical/pilot_joined`) additionally loads the two completed pilot runs and holds the answer-share, lexical-profile, association, and controlled-regression tables of Section 7, including every cell recorded as unidentified rather than fitted.

The companion bibliography records primary publication URLs/DOIs checked on 26 October 2026. Related-work claims are bounded to the corresponding task and language scope. In particular, MGSM is not a Turkish benchmark, ParaRel is not a logical-ordering benchmark, and grammatical minimal-pair evaluations are not answer-preserving inference pairs. The complete original CSV and compact table exports remain available with their native stratum identifiers; the paper uses selected summaries for readability rather than treating all dependent rows as new evidence.

C.1 Reproducing the pilot runs

The pilot runs of Section 7 execute from the project root with the pinned environment (notably `tiktoken==0.12.0` from the vendored wheel, which the grid layer verifies; a mismatched version refuses to plan or execute). Plans are write-once JSON artifacts: the executor re-derives the plan from the registry and environment at run time and compares byte-for-byte, so any registry or environment change invalidates a stored plan rather than silently altering the cohort. Run identifiers are permanent; a failed run id is retained as provenance and a fresh id is used for the retry.

```
# Derive an immutable plan (dry run), then execute it under a new run id.
PYTHONPATH=src python -m inference pilot --model <model-id> \
    --dry-run --output logs/<plan>.json
PYTHONPATH=src python -m inference pilot --plan logs/<plan>.json \
    --execute --run-id <run-id>

# An ambiguous transport error halts the run; resuming requires an
# explicit logged resolution acknowledging duplicate risk.
PYTHONPATH=src python -m inference resolve --run-id <run-id> \
    --plan logs/<plan>.json --task-id <task-id> \
    --decision retry --reason "<reason>" --acknowledge-duplicate-risk

# Paired analysis, tables, and figures for a completed run.
PYTHONPATH=src python -m analysis --root . --run <run-id> \
    --output results/analysis/<run-id> --figures
```

The executed runs used the registered configurations `qwen25-05b-hf` (pinned immutable revision, local `transformers`, CPU/float32, greedy) and `openai-gpt4o-snapshot` (`gpt-4o-2024-08-06`, temperature zero), enabled by explicit user selection in `configs/models.yaml`; the registry never enables a model automatically. The API arm requires `OPENAI_API_KEY` in the environment, and on macOS python.org builds, `SSL_CERT_FILE` pointed at a certificate bundle. Raw request/response records, normalized records, checkpoint stores, resolutions, and the derived analysis tables for both runs are retained under `results/` and `logs/` under the run ids `pilot-qwen05b-main-001` and `pilot-gpt4o-main-003` (failed attempts `pilot-gpt4o-main-001`/`-002` are retained as provenance); the summary report is `reports/pilot/PILOT_RESULTS.md`. Runs above the declared call threshold additionally require the dataset-bound two-reviewer Turkish linguistic review receipt, and frozen-test execution requires an approved pilot; neither gate is bypassable from the command line.

C.2 Reproducing the supplemental quantized local evaluation

The supplemental arm of Section 7.7 runs against a local Ollama server with the pinned Q4_K_M models; weight digests are recorded in every run header and checked against `results/p1/expected_digests.json`. The runner reuses the audited message constructor, which re-verifies each frozen prompt's SHA-256 at run time, and the registered strict parser; results append to one JSONL per model, and a re-invocation skips completed prompts exactly, so interrupted runs resume without duplication.

```
# Build (write-once) and semantically validate the expanded manifest.
PYTHONPATH=src python3 -m evaluation.build_p1_manifest
PYTHONPATH=src python3 -m evaluation.semantic_validator \
    --manifest data/manifests/p1_expanded_manifest.jsonl

# Execute one arm (resumable; pause by creating the flag file).
PYTHONPATH=src python3 -m inference.ollama_runner \
    --manifest data/manifests/p1_expanded_manifest.jsonl \
    --model qwen2.5:14b --output results/p1/qwen2-5-14b.jsonl \
```

```
    --pause-file results/p1/PAUSE

# Integrity verification, audited-primitive statistics, and the figure.
PYTHONPATH=src python3 -m evaluation.p1_integrity_report \
    --manifest data/manifests/p1_expanded_manifest.jsonl \
    --results results/p1/qwen2-5-*b.jsonl \
    --expected-digests results/p1/expected_digests.json \
    --output reports/inference_integrity_report.md
PYTHONPATH=src python3 -m analysis.p1_statistics \
    --manifest data/manifests/p1_expanded_manifest.jsonl \
    --results results/p1/qwen2-5-*b.jsonl --output results/p1/analysis
python3 paper/figures/manuscript/generate_p1_figures.py
```

The executed arms used Ollama digests `a8b0c515…` (0.5B), `357c53fb…` (3B), `845dbda0…` (7B), and `7cdf5a01…` (14B); generation options (temperature 0, `top_k` 1, seed 42, `num_ctx` 4096, `num_predict` 32) and the `/api/chat` endpoint are echoed on every record. The 0.5B arm executed on an Apple M1 (16 GB); the larger arms executed on a rented A100 from the byte-identical bundle (`colab/p1_colab_bundle.zip`), with manifest and bundle SHA-256 values asserted before any call. The statistical outputs in `results/p1/analysis/`, the integrity report, and the analysis summary `reports/p1_statistical_analysis.md` are retained; this arm is supplemental and quantized, and its records never enter the registered-provider analysis tables.

C.3 Rebuilding the manuscript

The entry point is `paper/main.tex`; chapter-like units are plain `.tex` files under `paper/sections/`. Compilation uses XeLaTeX and BibTeX with a Unicode-capable font and requires no model API, model weights, or experiment execution. Quantitative figures regenerate from the retained CSVs with `paper/figures/manuscript/generate_figures.py`, which also writes the plot manifest. The prior Markdown snapshot and its existing exported values are preserved for provenance; this LaTeX manuscript is the new reading and editing entry point.